

Part Rename Script (MJ)

Script Name: Part_Rename_MJ.lua

Functionality

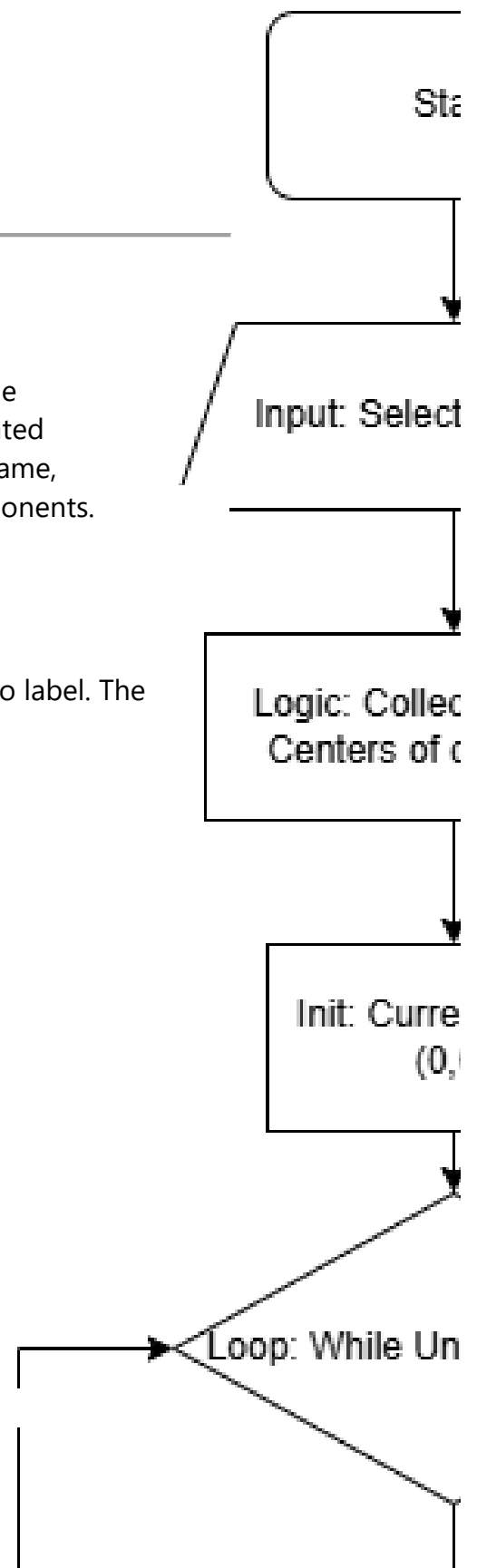
This script reorganizes the naming convention of parts in the active tray based on their spatial location. It applies a weighted linear cost function to sequentially label parts (e.g., [001]_Name, [002]_Name) in a path that minimizes jumps between components.

Logic Flow:

- **Initialization:** Starts from the origin (0,0).
- **Pathfinding:** It iteratively finds the "next" best part to label. The "Best" part is determined by minimizing a score:
 - $\text{Score} = (\text{Weight_Y} * |\text{Y_Dist}|) + (\text{Weight_Dist} * \text{Euclidean_Dist})$
- **Renaming:** The selected part is renamed with a sequential prefix (e.g., `[001]`), and the reference point moves to this part's center for the next iteration.

User Guide (Quick Start)

1. Open a tray with multiple positioned parts.
2. Run the script **Part_Rename_MJ**.
3. Select a folder for the log file.
4. The script will process the parts instantly.
5. Check the "Parts" list to see the new `[XXX]` prefixes.



Code Maintenance & Technical Notes

Weighted Logic: The script uses two tuning variables at the top of the file: ``Weight_Y_Distance`` (100.0) and ``Weight_Part_Distance`` (20.0). This configuration heavily penalizes Y-axis deviation (keeping the path in strips) while optimizing for general proximity.

Z-Height Note: While user notes mention Z-height logic, the current code implementation (as of 2026) calculates distances based on **2D X/Y coordinates**. The ``dist_sq`` function ignores the Z component. If 3D sorting is required, the distance function must be updated to include ``(p1.z - p2.z)^2``.

Safe Execution: The script wraps property access in ``pcall`` (protected call) to prevent crashes when accessing properties of potentially invalid mesh objects.